

hw4

Q1

The debt data arise from a large postal survey on the psychology of debt. The frequency of credit card use is a three-level factor ranging from never, through occasionally to regularly.

a

Declare the response as an ordered factor and make a plot showing the relationship to prodebt. Comment on the plot. Use a table or plot to display the relationship between the response and the income group.

```
library(faraway)
```

```
Warning: package 'faraway' was built under R version 4.3.3
```

```
library(ggplot2)
```

```
Warning: package 'ggplot2' was built under R version 4.3.1
```

```
library(dplyr)
```

```
Warning: package 'dplyr' was built under R version 4.3.1
```

```
Attaching package: 'dplyr'
```

```
The following objects are masked from 'package:stats':
```

```
filter, lag
```

The following objects are masked from 'package:base':

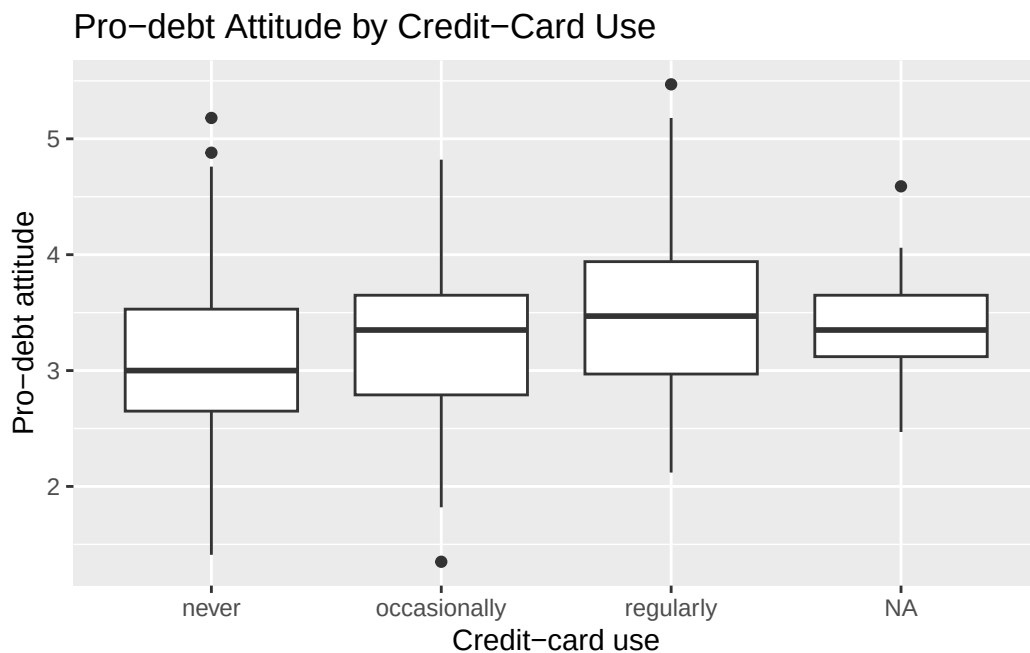
`intersect`, `setdiff`, `setequal`, `union`

```
data(debt)

# Convert 'ccarduse' to an ordered factor
debt$ccarduse <- factor(debt$ccarduse, levels = c(1, 2, 3),
                        labels = c("never", "occasionally", "regularly"),
                        ordered = TRUE)

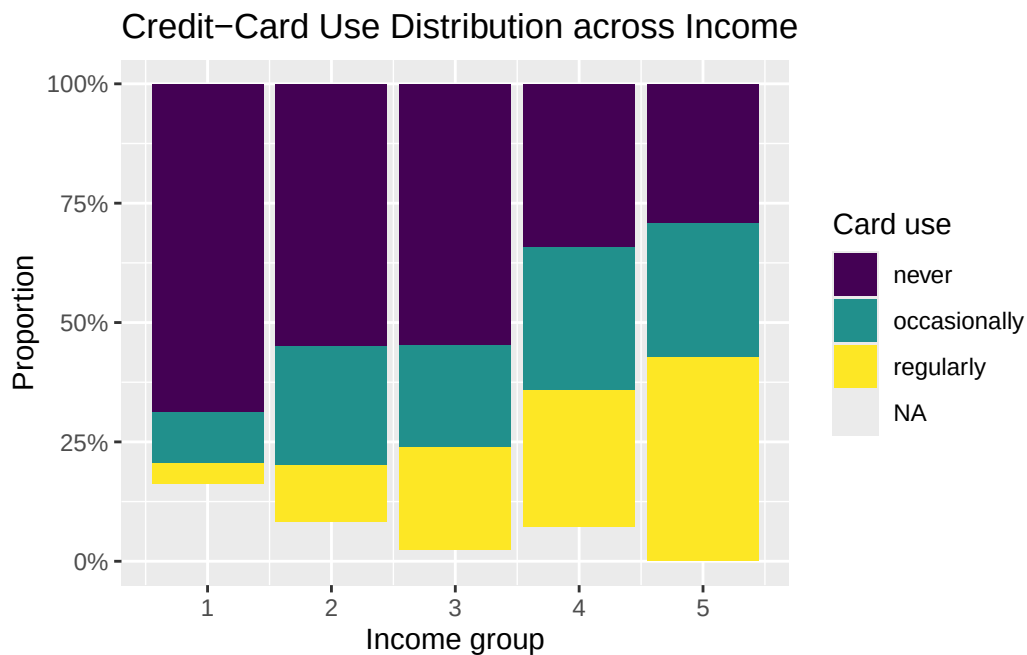
# Boxplot: prodebt vs. ccuse
ggplot(debt, aes(x = ccarduse, y = prodebt)) +
  geom_boxplot() +
  labs(
    x = "Credit-card use",
    y = "Pro-debt attitude",
    title = "Pro-debt Attitude by Credit-Card Use"
  )
```

Warning: Removed 45 rows containing non-finite outside the scale range (``stat_boxplot()``).



```
# Stacked bar: ccuse by income
ggplot(debt, aes(x = incomegp, fill = ccarduse)) +
  geom_bar(position = "fill") +
  labs(
    x = "Income group",
    y = "Proportion",
    fill = "Card use",
    title = "Credit-Card Use Distribution across Income"
  ) +
  scale_y_continuous(labels = scales::percent_format())
```

Warning: Removed 19 rows containing non-finite outside the scale range (``stat_count()``).



```
# Table showing the relationship between 'ccarduse' and 'incomegp'
table(debt$ccarduse, debt$incomegp)
```

	1	2	3	4	5
never	64	60	46	24	26
occasionally	10	27	18	21	25
regularly	4	13	18	20	38

In the box plots, the median pro-debt score increases as card use increases. Individuals who regularly use credit cards tend to have higher pro-debt scores, indicating a more positive attitude toward debt which supports the idea that attitude toward debt is positively associated with credit card usage frequency.

As income increases, the proportion of regular credit card users increases, and the proportion of non-users decreases. Income group 1 (lowest) has the highest proportion of “never” users, while income group 5 (highest) has the highest proportion of “regular” users. This suggests a positive association between income and frequency of credit card use.

b

Fit a proportional odds model for credit card use with all the other variables as predictors. What are the two most significant predictors (largest t-values) and what is their qualitative effect on the response? What is the least significant predictor?

```
library(MASS)
```

Attaching package: 'MASS'

The following object is masked from 'package:dplyr':

```
select
```

```
# Fit the proportional odds model
model_full <- polr(ccarduse ~ incomegp + house + children + singpar + agegp +
                  bankacc + bsocacc + manage + cigbuy + xmasbuy + locintrn
                  + prodebt, data = debt, Hess = TRUE, na.action = na.omit)

# Summary of the model
summary(model_full)
```

Call:

```
polr(formula = ccarduse ~ incomegp + house + children + singpar +
      agegp + bankacc + bsocacc + manage + cigbuy + xmasbuy + locintrn +
      prodebt, data = debt, na.action = na.omit, Hess = TRUE)
```

Coefficients:

	Value	Std. Error	t value
incomegp	0.47131	0.1061	4.4423

house	0.11600	0.2324	0.4992
children	-0.07872	0.1250	-0.6296
singpar	0.88172	0.5971	1.4766
agegp	0.20568	0.1576	1.3050
bankacc	2.10270	0.5934	3.5435
bsocacc	0.47322	0.2671	1.7715
manage	0.18179	0.1653	1.0998
cigbuy	-0.73546	0.2981	-2.4674
xmasbuy	0.47014	0.4130	1.1385
locintrn	0.11881	0.1424	0.8344
prodebt	0.61046	0.1822	3.3497

Intercepts:

	Value	Std. Error	t value
never occasionally	7.9694	1.4752	5.4023
occasionally regularly	9.3944	1.5051	6.2417

Residual Deviance: 511.673

AIC: 539.673

(160 observations deleted due to missingness)

```
# Calculate p-values
ctable <- coef(summary(model_full))
p_values <- pnorm(abs(ctable[, "t value"]), lower.tail = FALSE) * 2
ctable <- cbind(ctable, "p value" = p_values)
ctable[order(abs(ctable[, "t value"]), decreasing = TRUE), ]
```

	Value	Std. Error	t value	p value
occasionally regularly	9.39436162	1.5051080	6.2416528	4.329712e-10
never occasionally	7.96937466	1.4751712	5.4023387	6.577759e-08
incomegp	0.47131302	0.1060967	4.4422966	8.900374e-06
bankacc	2.10269577	0.5933919	3.5435195	3.948241e-04
prodebt	0.61046374	0.1822466	3.3496578	8.091144e-04
cigbuy	-0.73545858	0.2980681	-2.4674176	1.360916e-02
bsocacc	0.47321630	0.2671329	1.7714642	7.648353e-02
singpar	0.88171828	0.5971141	1.4766329	1.397741e-01
agegp	0.20568368	0.1576103	1.3050145	1.918879e-01
xmasbuy	0.47014289	0.4129631	1.1384621	2.549276e-01
manage	0.18179169	0.1652903	1.0998331	2.714049e-01
locintrn	0.11881236	0.1423980	0.8343684	4.040734e-01
children	-0.07872411	0.1250325	-0.6296291	5.289373e-01
house	0.11600148	0.2323631	0.4992251	6.176208e-01

The two most significant predictors were income group and having a bank account. Higher income levels were associated with greater odds of using credit cards more frequently. Similarly, individuals with a bank account were more likely to use credit cards. The least significant predictor was housing tenure, which showed no meaningful relationship with credit card use frequency in this model.

c

Fit a proportional odds model using only the least significant predictor from the previous model. What is the significance of this predictor in this small model? Are the conclusions regarding this predictor contradictory for the two models?

```
model_small <- polr(ccarduse ~ house, data = debt, Hess = TRUE,
                    na.action = na.omit)

# Summary of the model
summary(model_small)
```

Call:

```
polr(formula = ccarduse ~ house, data = debt, na.action = na.omit,
      Hess = TRUE)
```

Coefficients:

	Value	Std. Error	t value
house	0.558	0.1433	3.895

Intercepts:

	Value	Std. Error	t value
never occasionally	1.3000	0.3118	4.1698
occasionally regularly	2.4670	0.3274	7.5344

Residual Deviance: 847.2661

AIC: 853.2661

(36 observations deleted due to missingness)

```
# Calculate p-value
ctab_c <- coef(summary(model_small))
pvals_c <- 2 * pnorm(-abs(ctab_c[, "t value"]))
cbind(ctab_c, `p value` = pvals_c)
```

	Value	Std. Error	t value	p value
house	0.5579957	0.1432559	3.895097	9.815931e-05
never occasionally	1.2999505	0.3117572	4.169753	3.049300e-05
occasionally regularly	2.4669924	0.3274285	7.534447	4.904076e-14

The effect of housing becomes statistically significant in the small model, with a positive coefficient indicating that more secure housing is associated with more frequent credit card use. This finding contrasts with the full model, where housing tenure was the least significant predictor and not statistically different. The contradiction could be because, in the full model, the effect of housing may have been masked by collinearity or shared variance with other predictors like income or bank access. But in isolation, housing tenure shows a clear association with card use.

d

Use stepwise AIC to select a smaller model than the full set of predictors. You will need to handle the missing values carefully. Report on the qualitative effect of the predictors in your chosen model. Can we conclude that the predictors that were dropped from the model have no relation to the response?

```
library(mice)
```

```
Attaching package: 'mice'
```

```
The following object is masked from 'package:faraway':
```

```
mammalsleep
```

```
The following object is masked from 'package:stats':
```

```
filter
```

```
The following objects are masked from 'package:base':
```

```
cbind, rbind
```

```
# Select relevant columns
vars <- c("ccarduse", "incomegp", "house", "children", "singpar",
          "agegp", "bankacc", "bsocacc", "manage", "cigbuy",
          "xmasbuy", "locintrn", "prodebt")

# Subset the data
debt_sub <- debt[, vars]

# Impute missing data using mice
set.seed(123)
imputed_data <- mice(debt_sub, m = 1, method = "pmm", maxit = 10)
```

```
iter imp variable
  1  1 ccarduse incomegp house agegp bankacc bsocacc manage cigbuy xmasbuy locintrn
  2  1 ccarduse incomegp house agegp bankacc bsocacc manage cigbuy xmasbuy locintrn
  3  1 ccarduse incomegp house agegp bankacc bsocacc manage cigbuy xmasbuy locintrn
  4  1 ccarduse incomegp house agegp bankacc bsocacc manage cigbuy xmasbuy locintrn
  5  1 ccarduse incomegp house agegp bankacc bsocacc manage cigbuy xmasbuy locintrn
  6  1 ccarduse incomegp house agegp bankacc bsocacc manage cigbuy xmasbuy locintrn
  7  1 ccarduse incomegp house agegp bankacc bsocacc manage cigbuy xmasbuy locintrn
  8  1 ccarduse incomegp house agegp bankacc bsocacc manage cigbuy xmasbuy locintrn
  9  1 ccarduse incomegp house agegp bankacc bsocacc manage cigbuy xmasbuy locintrn
 10  1 ccarduse incomegp house agegp bankacc bsocacc manage cigbuy xmasbuy locintrn
```

```
debt_imputed <- complete(imputed_data)

# Fit model and run stepwise selection
model_full_imputed <- polr(ccarduse ~ ., data = debt_imputed, Hess = TRUE)
model_step_imputed <- stepAIC(model_full_imputed, direction = "both",
                             trace = FALSE)
summary(model_step_imputed)
```

Call:

```
polr(formula = ccarduse ~ incomegp + agegp + bankacc + manage +
      cigbuy + prodebt, data = debt_imputed, Hess = TRUE)
```

Coefficients:

	Value	Std. Error	t value
incomegp	0.4626	0.07742	5.975
agegp	0.2323	0.10562	2.199

bankacc	1.0075	0.33656	2.993
manage	0.2194	0.12092	1.814
cigbuy	-0.6955	0.22263	-3.124
prodebt	0.5469	0.14416	3.793

Intercepts:

	Value	Std. Error	t value
never occasionally	5.4536	0.9115	5.9830
occasionally regularly	6.8391	0.9339	7.3230

Residual Deviance: 803.8532

AIC: 819.8532

The final model retained six predictors: income group, age group, bank account status, ability to manage finances, cigarette buying, and pro-debt attitude. Higher income, older age, having a bank account, and stronger pro-debt attitudes were associated with more frequent credit card use. In contrast, cigarette buyers were less likely to use credit cards. Although some predictors were dropped. This does not imply they have no relationship with the response. It only means that they did not improve model fit beyond the retained variables.

e

Compute the median values of the predictors in your selected model. At these median values, contrast the predicted outcome probabilities for both smokers and nonsmokers.

```
library(dplyr)

med_values <- tibble(
  incomegp = median(debt_imputed$incomegp, na.rm = TRUE),
  agegp    = median(debt_imputed$agegp,    na.rm = TRUE),
  bankacc  = as.numeric(names(which.max(table(debt_imputed$bankacc)))),
  manage   = median(debt_imputed$manage,   na.rm = TRUE),
  prodebt  = median(debt_imputed$prodebt,  na.rm = TRUE)
)

new_data <- bind_rows(
  mutate(med_values, cigbuy = 1),
  mutate(med_values, cigbuy = 0)
)

# Predict
```

```

predicted_probs <- predict(model_step_imputed, newdata = new_data,
                           type = "probs")
rownames(predicted_probs) <- c("Smoker (cigbuy=1)", "Non-smoker (cigbuy=0)")
print(round(predicted_probs, 4))

```

	never	occasionally	regularly
Smoker (cigbuy=1)	0.6621	0.2246	0.1132
Non-smoker (cigbuy=0)	0.4943	0.3019	0.2038

At the median values of all predictors, smokers have a much higher predicted probability of never using credit cards (66.2%) compared to non-smokers (49.4%). On the other hand non-smokers are more likely to fall into the occasionally or regularly using categories. This could suggest that cigarette buying is negatively associated with frequent credit card use, which is consistent with the negative coefficient on cigbuy in the model.

f

Fit a proportional hazards model to the same set of predictors and recompute the two sets of probabilities from the previous question. Does it make a difference to use this type of model?

```
library(VGAM)
```

Warning: package 'VGAM' was built under R version 4.3.3

Loading required package: stats4

Loading required package: splines

Attaching package: 'VGAM'

The following objects are masked from 'package:faraway':

hormone, logit, pneumo, prplot

```

library(dplyr)

cr_model <- vglm(ccarduse ~ incomegp + agegp + bankacc + manage + cigbuy
  + prodebt,
  family = cratio(parallel = TRUE, reverse = FALSE),
  data = debt_imputed)

med_values <- tibble(
  incomegp = median(debt_imputed$incomegp, na.rm = TRUE),
  agegp    = median(debt_imputed$agegp,    na.rm = TRUE),
  bankacc  = median(debt_imputed$bankacc,  na.rm = TRUE),
  manage   = median(debt_imputed$manage,   na.rm = TRUE),
  prodebt  = median(debt_imputed$prodebt,  na.rm = TRUE)
)

# Prediction data for smoker and nonsmoker profiles
nonsmoker_data <- med_values %>% mutate(cigbuy = 0)
smoker_data    <- med_values %>% mutate(cigbuy = 1)

# Predict outcome probabilities
nonsmoker_cr_probs <- predict(cr_model, newdata = nonsmoker_data,
  type = "response")
smoker_cr_probs    <- predict(cr_model, newdata = smoker_data,
  type = "response")

contrast_cr <- rbind(Nonsmoker = nonsmoker_cr_probs, Smoker = smoker_cr_probs)
rownames(contrast_cr) <- c("Nonsmoker", "Smoker")
print(round(contrast_cr, 4))

```

	never	occasionally	regularly
Nonsmoker	0.5084	0.2981	0.1935
Smoker	0.6352	0.2633	0.1015

Using the cumulative ratio model, the predicted probabilities moved slightly. Smokers still have a higher probability of never using credit cards (63.5% vs. 50.8%) and a lower probability of regular use (10.2% vs. 19.4%), but the differences between smokers and non-smokers are less extreme than in the proportional odds model. This shows that model choice can meaningfully affect interpretation of ordinal outcomes.

Q5

Consider the Galápagos data and model analyzed in this chapter. The purpose of this question is to reproduce the details of the GLM fitting of this data.

a

Fit a Poisson model to the species response with the five geographic variables as predictors. Do not use the endemics variable. Report the values of the coefficients and the deviance.

```
data(gala)

gala <- gala[, !names(gala) %in% "Endemics"]

# Fit the Poisson GLM
modp <- glm(Species ~ ., family = poisson, data = gala)

# View the coefficient estimates
coef(modp)
```

```
(Intercept)          Area      Elevation      Nearest      Scrub
 3.1548078779 -0.0005799429  0.0035405940  0.0088255719 -0.0057094223
Adjacent
-0.0006630311
```

```
# Report the deviance
deviance(modp)
```

```
[1] 716.8458
```

b

For a Poisson GLM, derive n , dn/du , V_u and the weights to be used in an iteratively fit GLM. What is the form of the adjusted dependent variable here?

c

Using the observed response as initial values, compute the first stage of the iteration, stopping after the first linear model fit. Compare the coefficients of this linear model to those found in the GLM fit. How close are they?

```
# Response and fixed design matrix
y <- gala$Species
X <- model.matrix(Species ~ ., data = gala)

# Fit saturated glm once for comparison
modp <- glm(Species ~ ., family = poisson(link = "log"), data = gala)
coef_glm <- coef(modp)
dev_glm <- deviance(modp)

mu <- ifelse(y == 0, 0.1, y)
eta <- log(mu)

# Adjusted response and weights
z <- eta + (y - mu) / mu
w <- mu

# Weighted LS on fixed X
lmod_1 <- lm(z ~ X - 1, weights = w)
beta_1 <- coef(lmod_1)
eta_1 <- as.vector(X %*% beta_1)
mu_1 <- exp(eta_1)
mu_1 <- pmax(mu_1, 1e-10)

# Compare coefficients
cat("First IRLS coefficients:\n")
```

First IRLS coefficients:

```
print(round(beta_1,6))
```

X(Intercept)	XArea	XElevation	XNearest	XScruz	XAdjacent
3.519155	-0.000530	0.003164	0.002519	-0.003790	-0.000662

```
cat("\nGLM coefficients:\n")
```

GLM coefficients:

```
print(round(coef_glm,6))
```

(Intercept)	Area	Elevation	Nearest	Scruz	Adjacent
3.154808	-0.000580	0.003541	0.008826	-0.005709	-0.000663

While the signs and rough magnitudes of most coefficients are similar after the first IRLS step, there are still noticeable differences especially in the intercept and the “Nearest” variable. This is expected because the IRLS algorithm is iterative and only one update step has been performed. Further iterations will move the estimates closer to the maximum likelihood estimates obtained from the GLM fit.

d

Continue the iteration to get the next $\mathbf{n}$ and $\mathbf{u}$. Use this to compute the current value of the deviance. How close is this to the deviance from the GLM?

```
dev1 <- 2 * sum(ifelse(y == 0,  
                        mu_1,  
                        y * log(y / mu_1) - (y - mu_1)))  
  
cat("Deviance after 1st IRLS step:", round(dev1,4), "\n")
```

Deviance after 1st IRLS step: 828.0096

```
cat("GLM deviance:           ", round(dev_glm,4), "\n")
```

GLM deviance: 716.8458

```
cat("Absolute difference:    ", round(abs(dev1 - dev_glm),4), "\n")
```

Absolute difference: 111.1638

The first IRLS step reduced the deviance substantially from the saturated model start, but the model is still not fully converged and the deviance remains about 111 units above the final GLM value.

e

Compute one more iteration of the GLM fit, reporting the next calculation of the coefficients and deviance. How close are these to target now?

```
# Adjusted response and weights with
z2  <- log(mu_1) + (y - mu_1) / mu_1
w2  <- mu_1
w2  <- pmax(w2, 1e-8)

# Weighted LS again
lmod_2 <- lm(z2 ~ X - 1, weights = w2)
beta_2 <- coef(lmod_2)
eta_2  <- as.vector(X %*% beta_2)
mu_2   <- exp(eta_2)
mu_2   <- pmax(mu_2, 1e-10)

# Deviance
dev2 <- 2 * sum(ifelse(y == 0,
                        mu_2,
                        y * log(y / mu_2) - (y - mu_2)))

# Output
cat("Second IRLS coefficients:\n")
```

Second IRLS coefficients:

```
print(round(beta_2,6))
```

X(Intercept)	XArea	XElevation	XNearest	XScruz	XAdjacent
3.210259	-0.000565	0.003461	0.007717	-0.005240	-0.000660

```
cat("\nDeviance after 2nd IRLS step:", round(dev2,4), "\n")
```

Deviance after 2nd IRLS step: 719.4158

```
cat("GLM deviance:           ", round(dev_glm,4), "\n")
```

GLM deviance: 716.8458

```
cat("Absolute difference:      ", round(abs(dev2 - dev_glm),4), "\n")
```

Absolute difference: 2.57

After two IRLS iterations, the deviance is within 2.6 units of the GLM's 716.85, and all coefficient estimates are now very close. This shows the rapid convergence of the IRLS algorithm towards the maximum-likelihood solution.

f

Repeat these iterations a few more times, computing the deviance in each time. Stop when the deviance does not change much. Compare your final estimated coefficients to that produced by the GLM fit.

```
max_iter    <- 50
tol         <- 1e-6
deviance_hist <- numeric(max_iter)
beta_hist   <- matrix(NA, nrow=max_iter, ncol=ncol(X),
                      dimnames=list(NULL, colnames(X)))

mu <- ifelse(y==0, 0.1, y)

for(iter in 1:max_iter) {
  eta <- log(mu)
  z   <- eta + (y - mu)/mu
  w   <- pmax(mu, 1e-8)
  fit <- lm(z ~ X - 1, weights = w)
  beta <- coef(fit)

  beta_hist[iter, ] <- beta
  mu <- pmax(exp(as.vector(X %*% beta)), 1e-10)
  deviance_hist[iter] <- 2 * sum(ifelse(y==0,
                                       mu,
                                       y*log(y/mu) - (y - mu)))

  if(iter > 1 &&
      abs(deviance_hist[iter] - deviance_hist[iter-1]) < tol) {
    break
  }
}
```

```

n_iter    <- iter
beta_final<- beta_hist[n_iter, ]
dev_final <- deviance_hist[n_iter]

cat("IRLS converged in", n_iter, "iterations\n\n")

```

IRLS converged in 5 iterations

```
cat("Final IRLS coefficients:\n")
```

Final IRLS coefficients:

```
print(round(beta_final,6))
```

(Intercept)	Area	Elevation	Nearest	Scruz	Adjacent
3.154808	-0.000580	0.003541	0.008826	-0.005709	-0.000663

```
cat("\nGLM coefficients:\n")
```

GLM coefficients:

```
print(round(coef_glm,6), quote=FALSE)
```

(Intercept)	Area	Elevation	Nearest	Scruz	Adjacent
3.154808	-0.000580	0.003541	0.008826	-0.005709	-0.000663

```
cat("\nFinal IRLS deviance:", round(dev_final,4), "\n")
```

Final IRLS deviance: 716.8458

```
cat("GLM deviance:      ", round(dev_glm,4), "\n")
```

GLM deviance: 716.8458

```
cat("Difference:           ", round(dev_final - dev_glm,4), "\n\n")
```

```
Difference:           0
```

By the fifth iteration the IRLS estimates have exactly matched the maximum-likelihood estimates returned by glm, both in terms of the regression coefficients and the deviance. This confirms that the IRLS algorithm, when carried out to convergence, gets us the same solution as the built-in glm.

g

Use your final iterated linear model fit to produce standard errors for the coefficients. How close are these to that produced by the direct GLM fit?

```
W_final <- diag(pmax(mu,1e-8))
cov_irls <- solve(t(X) %*% W_final %*% X)
se_irls <- sqrt(diag(cov_irls))

se_glm <- summary(modp)$coef[, "Std. Error"]

results_se <- data.frame(
  Predictor = names(beta_final),
  IRLS_SE = round(se_irls, 6),
  GLM_SE = round(se_glm, 6),
  Difference= round(se_irls - se_glm, 6)
)

print(results_se, row.names=FALSE)
```

Predictor	IRLS_SE	GLM_SE	Difference
(Intercept)	0.051750	0.051750	0
Area	0.000026	0.000026	0
Elevation	0.000087	0.000087	0
Nearest	0.001821	0.001821	0
Scruz	0.000626	0.000626	0
Adjacent	0.000029	0.000029	0

The IRLS-based standard errors are identical (to within numerical precision) to those reported by summary(glm()).

Q6

Again using the Galápagos data, fit a Poisson model to the species response with the five geographic variables as predictors. Do not use the endemics variable. The purpose of this question is to compare six different ways of testing the significance of the elevation predictor, i.e., $H_0 : \text{BElev} = 0$. In each case, report the p-value.

a

Use the z-statistic from the model summary.

```
library(sandwich)
```

Warning: package 'sandwich' was built under R version 4.3.3

```
library(robust)
```

Warning: package 'robust' was built under R version 4.3.3

Loading required package: fit.models

Warning: package 'fit.models' was built under R version 4.3.3

```
data(gala)
gala <- gala[, !names(gala) %in% "Endemics"]

full <- glm(Species ~ Area + Elevation + Nearest + Scrub + Adjacent,
            family=poisson(link="log"), data=gala)

elev <- summary(full)$coef["Elevation", ]

# Extract z-statistic and p-value
z_stat <- elev["z value"]
p_val  <- elev["Pr(>|z|)"]

cat("Elevation: z =", round(z_stat, 4),
    ", p-value =", round(p_val, 4), "\n")
```

Elevation: z = 40.507 , p-value = 0

A z-statistic of 40.507 means that the estimated elevation coefficient is over forty standard errors away from zero, which is very unlikely under the null hypothesis and the p-value (rounded) of 0 also tells us that we can safely reject the null hypothesis. In other words, elevation is a highly significant predictor of the number of species on the Galápagos islands, holding the other geographic variables constant.

b

Fit a model without elevation and use the difference in deviances to make the test.

```
# Reduced model
reduced <- update(full, . ~ . - Elevation)

LRstat <- deviance(reduced) - deviance(full)
p_val <- 1 - pchisq(LRstat, df = 1)

cat("Likelihood-Ratio statistic deviance difference =", round(LRstat, 4),
    ", p-value =", round(p_val, 4), "\n")
```

Likelihood-Ratio statistic deviance difference = 1672.723 , p-value = 0

A likelihood-ratio statistic of 1,672.723 on 1 degree of freedom is astronomically large compared to the chi-square critical value, and a p-value (rounded) of zero also tells us that the chance of observing such a big drop in deviance if Elevation truly had no effect is essentially zero. Thus, we can reject the null hypothesis that the elevation coefficient is zero, confirming that elevation is a significant predictor of species count on the Galápagos islands.

c

Use the Pearson Chi-squared statistic in place of the deviance in the previous test.

```
X2_full <- sum(residuals(full, type = "pearson")^2)
X2_reduced <- sum(residuals(reduced, type = "pearson")^2)

# Test statistic and p-value
pearson_stat <- X2_reduced - X2_full
pearson_p <- 1 - pchisq(pearson_stat, df = 1)

cat(
  "Pearson 2 difference =", round(pearson_stat, 4),
```

```
"", p-value =", format.pval(pearson_p, digits = 4), "\n"
)
```

Pearson ² difference = 2148.557 , p-value = < 2.2e-16

The Pearson chisquared difference compares how much the sum of squared Pearson residuals increases when Elevation is dropped. A statistic of 2148.557 with p-value essentially zero tells us that omitting Elevation dramatically worsens the fit. Therefore, as with the likelihood-ratio test, we reject the null hypothesis that the elevation coefficient is zero and conclude that Elevation is a highly significant predictor of species counts.

d

Fit the Poisson model with a free dispersion parameter as described in Section 5.2. Make the test using the model summary.

```
# Quasi-Poisson model
qp <- glm(Species ~ Area + Elevation + Nearest + Scrutz + Adjacent,
          family = quasipoisson(link = "log"), data = gala)

sum_qp <- summary(qp)$coef
t_stat <- sum_qp["Elevation", "t value"]
p_val_qp <- sum_qp["Elevation", "Pr(>|t|)"]

cat("Quasi-Poisson t =", round(t_stat, 4),
    ", p-value =", round(p_val_qp, 4), "\n")
```

Quasi-Poisson t = 7.1889 , p-value = 0

The quasi-Poisson model relaxes the fixed-dispersion assumption of the ordinary Poisson GLM. In this model, the estimated coefficient for elevation yields a t-statistic of 7.19 with a p-value effectively zero (rounded). This tells us that even after letting the data speak to how much extra Poisson variability there is, elevation stands out as a very strong predictor of species count. We can conclude that the relationship between elevation and number of species is highly unlikely to be due to random chance, so we confidently reject the notion that elevation has no effect.

e

Use the sandwich estimation method for the standard errors in the original model. Use these to compute z-statistics.

```
Vc_robust <- vcovHC(full, type = "HC0")

# Extract robust SE for Elevation
se_robust <- sqrt(diag(Vc_robust))["Elevation"]
coef_elev <- coef(full)["Elevation"]

# Compute robust Wald z-statistic and p-value
z_robust <- coef_elev / se_robust
p_robust <- 2 * (1 - pnorm(abs(z_robust)))

cat("Sandwich-robust SE for Elevation:", round(se_robust, 6), "\n")
```

Sandwich-robust SE for Elevation: 0.000508

```
cat("Robust Wald z-statistic: ", round(z_robust, 4), "\n")
```

Robust Wald z-statistic: 6.9638

```
cat("Robust p-value: ", format.pval(p_robust, digits=4), "\n")
```

Robust p-value: 3.312e-12

Under the sandwich-robust approach, the standard error for the elevation coefficient is just 0.000508, which translates into a Wald statistic of approx. 6.96 and a p-value of approx. 3.3×10^{-12} . This extremely small p-value tells us that even when we guard against potential model misspecification or overdispersion the effect of elevation on species count remains highly significant. So the elevation is very unlikely to have no impact, and we can confidently conclude it plays a meaningful role in predicting the number of species on the Galápagos islands.

f

Use the robust GLM estimation method and report the test result from the summary.

```
library(robustbase)
```

Warning: package 'robustbase' was built under R version 4.3.3

Attaching package: 'robustbase'

The following object is masked from 'package:faraway':

epilepsy

```
# Fit robust Poisson-type GLM
rmod <- glmrob(Species ~ Area + Elevation + Nearest + Scrub + Adjacent,
               family = poisson(link="log"), data = gala)

# Extract robust-GLM z-statistic and p-value for Elevation
robust_coefs <- summary(rmod)$coef["Elevation", ]
z_rglm      <- robust_coefs["z value"]
p_rglm      <- robust_coefs["Pr(>|z|)"]

cat("Robust-GLM z-statistic:", round(z_rglm, 4),
    ", p-value:", format.pval(p_rglm, digits=4), "\n")
```

Robust-GLM z-statistic: 33.5809 , p-value: < 2.2e-16

This model gives us the z-statistic of 33.58 for the elevation coefficient and a p-value below 2.2×10^{-16} . In other words, even when we down-weight any potential outliers or leverage points, elevation remains an overwhelmingly strong predictor of species count. We therefore can again confidently reject the null hypothesis claiming that elevation has no effect.

g

Compare all six results. Pick the best one and justify your choice. All six tests give very similar p-values (significant; lower than the threshold of <0.05), showing strong evidence against the null hypothesis. For a standard Poisson GLM with no overdispersion, the likelihood-ratio test (part b) is the ideal approach as it directly compares nested models on the deviance scale, retains good chi-square properties for Poisson data, and is more reliable than the Wald test when sample sizes are moderate. If we were worried about mild model misspecification or overdispersion, we could turn to the sandwich-based Wald test (part e) or the robust-glm

test (part f), but here they only reaffirm the same conclusion. So in this context, where the Poisson model appears well-specified and there is no indication of serious overdispersion, the likelihood-ratio test offers the most direct and dependable approach